PETEAT: A WEB & MOBILE PLATFORM FOR VETERINARY CLINICS AND PET LOVERS WITH NFC TRACKING, TELEMEDICINE, AND CLOUD BASED ELECTRONIC MEDICAL RECORDS

by

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ABSTRACT

This study presents PetEat, an innovative web and mobile platform designed to revolutionize veterinary care and pet ownership through the integration of Near Field Communication (NFC) tracking, telemedicine, and cloud-based Electronic Medical Records (EMR). Addressing critical challenges such as fragmented record-keeping, inefficient appointment scheduling, and the distress of lost pets, PetEat offers a unified solution to enhance accessibility, operational efficiency, and overall pet well-being. The platform aims to connect veterinary clinics and pet owners, facilitate remote consultations, ensure secure and timely healthcare management with automated reminders, and improve lost pet recovery through NFC technology.

Employing an Agile development methodology, the research followed a structured approach including requirements analysis, system design, and rigorous testing. The system architecture leverages web and mobile applications with Next.js, Flutterflow, Node.js, Firebase, and Socket.io to ensure scalability and robust functionality. A comprehensive survey was conducted among IT Experts (n=10) and End Users (Veterinarians and Pet Owners, n=30) to evaluate the platform's functional suitability, performance efficiency, and usability based on ISO 25010:2011 standards.

The results demonstrate overwhelmingly positive feedback. Key findings indicate that the platform excels in Functional Completeness (IT Experts: 3.9, End Users: 4.1), User Error Protection (IT Experts: 4.4, End Users: 4.1), and

Functional Appropriateness (IT Experts: 4.4, End Users: 4.43), all interpreted as "Agree." Both groups consistently rated the system highly, validating its core functionalities, user-friendliness, and reliability. While areas like Operability (IT Experts: 4.1, End Users: 3.9) and Functional Completeness (IT Experts: 3.9, End Users: 4.1) received slightly lower, though still positive, ratings, the study concludes that PetEat provides a valuable and much-needed technological advancement in the pet care industry. Recommendations for future enhancements include implementing payment gateway integration, Predictive analytics, multi-language support, RFID, and further performance optimization to cater to a broader audience and ensure long-term sustainability.

Keywords: *NFC Tracking, Pets, Telemedicine, Veterinary Clinics, Web and Mobile*

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LIST OF ABBREVIATIONS

API - Application Programming Interfaces
AWS - Amazon Web Services
CAT - Citizen Army Training
COVID-19 - Coronavirus Disease 2019
CPU - Central Processing Unit
DFD - Data Flow Diagram
EMR - Electronic Medical Record
ERD - Entity Relationship Diagram
FEU - Far Eastern University
GB - Gigabyte
GPU - Graphics Processing Unit
ID - Identification
IT - Information Technology
IPO - Input-Process-Output
ISO - International Organization for Standardization
NFC - Near Field Communication
OS - Operating System
QR - Quick Response
RAM - Random Access Memory
UAT - User Acceptance Testing
UML - Unified Modeling Language

Chapter 1

INTRODUCTION

1.1 Background of the Study

The pet care industry is evolving rapidly, driven by technological advancements and an increasing demand for integrated solutions in animal health monitoring, communication, and convenience for both veterinary clinics and pet owners. The global pet care market continues to expand, reflecting the growing importance of pets in households worldwide (Grand View Research, 2023). In the Philippines, pet ownership is on the rise, contributing to a higher demand for veterinary services and pet-related monitoring products (Philippine Statistics Authority, 2023). This growth underscores the need for a platform that addresses the existing gaps in veterinary care and pet management.

Currently, veterinary clinics face challenges due to disjointed systems for managing records, appointments, and communication. This fragmentation affects operational efficiency and leads to inconsistencies in patient care (Veterinary Practice News, 2023). Pet owners often struggle to access critical health information and professional support, impacting their ability to make informed decisions about their pets' well-being. An emerging concern is the lack of effective tracking solutions of lost pets, especially regarding pet owners information (e.g. pet owner name and contact number). In the United States alone, approximately 10 million pets are lost each year, with only a fraction reunited with their owners (American Pet Products Association, 2023). Near Field Communication (NFC) technology enhances pet identification by enabling quick, contactless retrieval of pet information through smartphone scanning,

improving lost pet recovery and veterinary access (QRLAB, 2023). Pet monitoring and veterinary clinic management continue to face significant challenges, particularly in lost pet identification, medical record-keeping, inventory management. To validate these issues, an interview and survey were conducted with a veterinarian at Pet Express in Festival Mall Alabang, Muntinlupa. The findings revealed that they still rely on manual methods for medical records and inventory management, leading to inefficiencies. The veterinarian confirmed that transitioning to a cloud-based EMR would improve accessibility. Additionally, manual inventory management often results in shortages and inefficiencies. Implementing an inventory management system would address these issues and ensure accurate tracking. The veterinarian also recognized the value of telemedicine for remote video consultations, allowing pet owners to save time, reduce costs, and access veterinary care more conveniently

For lost pet identification, the doctor supported Near Field Communication (NFC) technology, which enables quick, contactless pet identification through smartphone scanning. Unlike microchips, NFC tags allow immediate owner retrieval without a vet visit. However, NFC is not real-time tracking and relies on public scanning. NFC technology is increasingly being utilized in pet identification due to its efficiency and ease of use. NFC-enabled pet tags allow individuals to access a pet's profile by simply tapping the tag with an NFC-capable device, providing immediate access to the owner's community contact information and the pet's medical history. This technology offers a modern alternative to traditional identification methods, enhancing the likelihood of reuniting lost pets with their owners. Unlike microchips, which require a visit to a veterinarian for scanning, NFC tags can be read by any smartphone with

NFC capabilities, facilitating quicker and more accessible identification. However, it's important to note that NFC tags do not provide real-time tracking; they rely on someone finding the pet and scanning the tag to access the stored information (QR Lab, 2024).

To address these issues, the researchers propose PetEat, a unified web and mobile platform for telemedicine, inventory management, and cloud-based EMR and the integration of NFC technology, community-driven. This platform aims to enhance the veterinary care experience and improve pet ownership outcomes by:

1.2 Objectives of the Study

General Objective

To design and develop PetEat, a Web and Mobile platform that enhances veterinary care and pet ownership experiences by integrating Telemedicine, NFC Technology, Cloud-based Electronic Medical Record (EMR), and Inventory Management.

Specific Objectives

1. To develop a web and mobile application to establish connections between veterinary clinics and pet owners to enhance veterinary care accessibility and operational efficiency.
2. To overcome the challenges of in-person veterinary visits by adopting telemedicine to enable remote video consultations with veterinarians, ensuring timely care while saving time and costs.

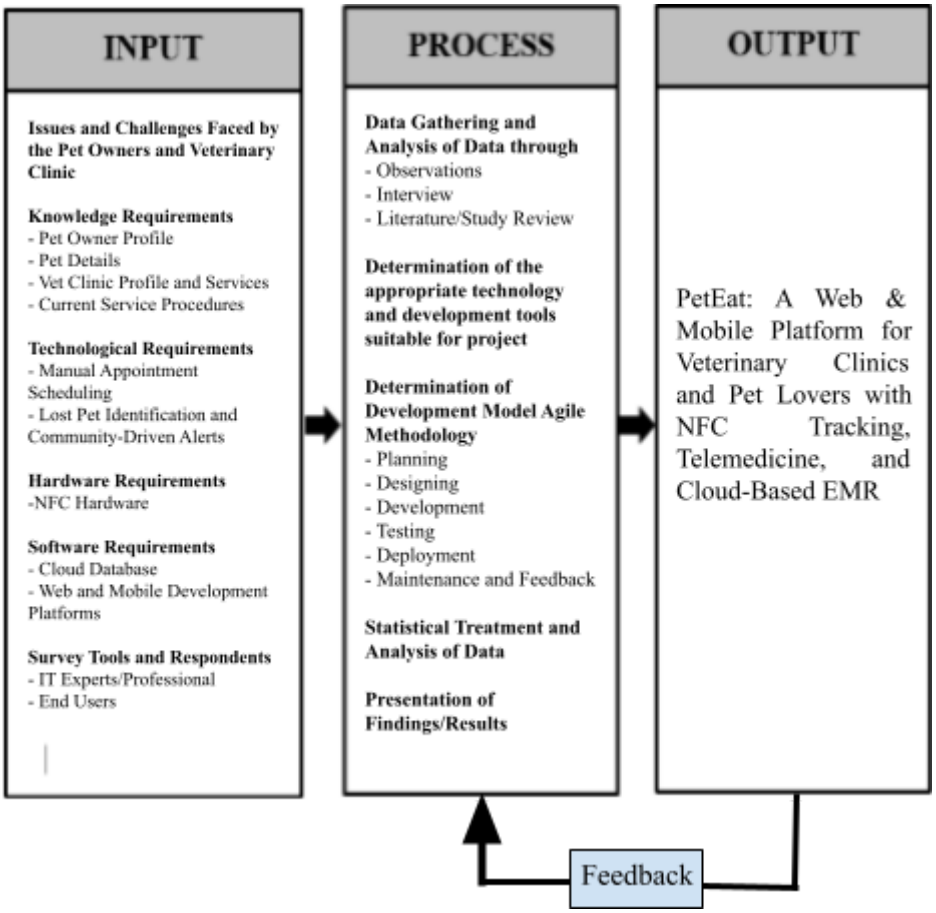
3. To replace traditional manual record-keeping by utilizing Cloud-Based EMR to ensure secure storage of medical records and timely healthcare management with automated reminders for pet vaccinations and check-ups.
4. To address the issue of lost pets by utilizing NFC technology for quick identification and community-driven lost pet reporting, enhancing recovery and safety.
5. To mitigate inventory shortages and inefficiencies by integrating an inventory management system to enhance accurate tracking.

1.3 Conceptual Framework of the Study

The conceptual framework integrates cutting-edge technologies such as NFC, telemedicine, and cloud-based EMRs within a user-centric design. These components work together to address existing challenges in veterinary care and create an efficient and accessible platform for users.

Figure 1.1

Input- Process- Output Diagram



This IPO (Input-Process-Output) diagram outlines the development of PetEat, a web and mobile platform for veterinary clinics and pet owners. The input phase encompasses the identification of issues faced by pet owners and clinics, the gathering of knowledge requirements (pet/clinic profiles, service procedures), technological requirements (NFC, appointment scheduling), hardware (NFC hardware), software (cloud database, development platforms), and survey data from IT experts and end-users. The process involves rigorous data collection and analysis through observations, interviews, and literature reviews, followed by the determination of suitable technology, development tools, and an Agile development model. This leads to

the systematic planning, design, development, testing, deployment, and maintenance of the platform, culminating in statistical analysis and presentation of findings. The output is the fully realized PetEat platform, offering NFC tracking, telemedicine, and Cloud-Based EMR. Crucially, feedback loops from the output stage inform subsequent iterations, ensuring continuous improvement and user satisfaction.

1.4 Significance of the Study

This study is significant as it aims to address major challenges in pet healthcare and veterinary clinic management by leveraging technology. By integrating various digital tools into a single platform, PetEat enhances communication, improves efficiency, and provides better access to pet healthcare services. The following groups will directly benefit from the study.

Pet Owners. PetEat will provide pet owners with enhanced access to veterinary care, telemedicine consultations, medical records, and pet tracking features.

Veterinarians. PetEat will benefit veterinarians from improved operational workflows, centralized medical records, and a streamlined appointment and inventory management system.

Pet shelters and rescue organizations. PetEat will be able to provide tools to maintain pet health records and use NFC tag scanning to assist in identifying lost pets and reconnecting them with their owners.

Future Software Developers. PetEat will be able to provide references for developing integrated pet care platforms.

1.5 Scope and Delimitations

The PetEat platform will be tested by selected veterinary clinics within Alabang, Muntinlupa, including Pet Express, to provide a solution for veterinary care and pet ownership. It includes telemedicine functionality for consultations, enabling pet owners to seek professional advice remotely. However, telemedicine requires a stable internet connection for seamless communication and is dependent on the availability of veterinary clinics' schedules. Additionally, clinics must transition to digital records, which may face resistance from traditional practices. The system incorporates a cloud-based Electronic Medical Record (EMR) feature for secure and accessible storage of pet medical histories.

For pet identification, NFC technology is utilized to facilitate quick pet verification when scanned with an NFC-enabled device. Since NFC has a short-range detection capability (approximately 5 cm), pet identification requires close proximity scanning. As an alternative, a Quick Response (QR) code is also embedded on the pet's tag, allowing anyone with a smartphone camera to scan it and retrieve the owner's details. This ensures that lost pets can still be identified even if the finder's device does not support NFC.

Veterinary clinics can benefit from inventory management tools that make tracking supplies easier and reduce stock issues. However, it has some limitations, such as possible data inaccuracy due to manual input errors, integration challenges with other features like telemedicine and cloud-based EMR, and limited automati**on**, which may cause delays in restocking supplies.

The platform is exclusively designed for Android OS, as iOS is not covered in

this study. However, users can also access the system through a web, ensuring availability on both desktop and mobile devices. For the best experience, Google Chrome is the recommended browser for both platforms. Moreover, the system primarily focuses on common household pets such as dogs and cats, excluding other types of animals. The effectiveness of NFC tracking depends on active user participation, which is more viable in urban areas.

1.6 Definition of Terms

The following terms used by the researchers are significant in deeply understanding the study. Terms are defined both functionally and lexically.

Operational Terms

Appointment Scheduling. A feature allowing pet owners to book and manage veterinary appointments seamlessly.

EMR (Electronic Medical Records). An Electronic Medical Records system to securely store information on the health of the pet.

NFC Tracking. Technology to locate and retrieve lost pets with the use of NFC tags.

Pet Care Industry. Refers to businesses and services related to the health, nutrition, grooming, and overall well-being of pets.

Pet Owners. Individuals who own and are responsible for the care of pets.

Pet Profile Management. Functionality that enables users to manage information for multiple pets within a single account.

Telemedicine. Video-based consultations allowing remote diagnosis.

Vaccination Reminders. Automated alerts sent to pet owners to ensure timely health care interventions.

Veterinary Clinics. Establishments providing medical services for pets, including health check-ups, vaccinations, and treatments.

Technical Terms

API Integration. The process of connecting two or more software systems using Application Programming Interfaces (APIs) to enable seamless data exchange and communication (Postman, 2021).

Bluetooth. A wireless technology used in some pet tracking devices with short-range communication capability (Wikipedia, n.d.).

Cloud Computing. A technology that provides computing services such as data storage, processing, and networking over the internet, reducing the reliance on local hardware and infrastructure (GeeksforGeeks, 2024).

Compatibility. Degree to which a product, system, or component can exchange information with other products, systems, or components, and/or perform its required functions, while sharing the same hardware or software environment (ISO/IEC 25010:2011(en).

Data Encryption. The process of converting readable data (plaintext) into an unreadable format (ciphertext) to protect sensitive information from unauthorized access (GeeksforGeeks, 2025).

Disjointed Systems. Separate or unintegrated software systems that complicate management and communication processes in veterinary clinics (Prime Penguin, n.d.).

Functional Suitability. Degree to which a product or system provides functions that meet the specified requirements (ISO/IEC 25010:2011(en).

Inventory Management. A system to monitor and control the stock of medical supplies and pet-related products in veterinary clinics (Investopedia, 2024).

Maintainability. Degree of effectiveness and efficiency with which a product or system can be modified by the intended maintainers (ISO/IEC 25010:2011(en).

Mobile-Responsive Design. A design approach that ensures the platform functions optimally on various device types, including smartphones and tablets (LambdaTest, 2023).

NFC (Near Field Communication). A short-range wireless technology enabling communication between devices when they are close to each other, used in PetEat for location tracking (Investopedia, 2023).

Performance Efficiency. Performance relative to the amount of resources that can include other software products, the software and hardware configuration of the system, and materials (e.g. print paper, storage media) (ISO/IEC 25010:2011(en).

Portability. Degree of effectiveness and efficiency with which a system, product, or component can be transferred from one hardware, software, or other operational or usage environment to another (ISO/IEC 25010:2011(en).

Reliability. Degree to which a system, product, or component performs specific functions under specified conditions for a specified period of time (ISO/IEC 25010:2011(en).

Security. Degree to which a product or system protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization (ISO/IEC 25010:2011(en).

Telemedicine. The practice of providing medical consultations, diagnoses, and treatments remotely using digital communication tools, reducing the need for in-person visits (Medical News Today, 2023).

User Authentication. A security feature ensuring that only authorized users can access the platform (GeeksforGeeks, 2024).

Usability. Degree to which a product or system can be used by specified users to achieve specific goals with effectiveness, efficiency, and satisfaction in a specified context of use (ISO/IEC 25010:2011(en).

Web and Mobile Platform. A web and mobile platform refers to software applications that operate on both web browsers and mobile devices, enabling users to access services from multiple interfaces (BuildFire, 2024).

Chapter 2

REVIEW OF RELATED LITERATURE

The rise of digital technologies has transformed veterinary care by addressing challenges in accessibility and efficiency. In the Philippines, where pet ownership is growing rapidly, the need for centralized platforms like PetEat has become evident. This chapter explores literature supporting the integration of telemedicine, NFC tracking, and Cloud Based EMR to enhance pet healthcare and management.

2.1 Local Literature

Recent studies in the Philippines have highlighted the growing role of technology in improving pet healthcare and management. The Philippine Veterinary Medical Association (2023) reported that over 70% of veterinary services are concentrated in urban regions, emphasizing the need for accessible platforms to extend care to rural areas. This aligns with PetEat's goal of bridging geographical gaps through telemedicine and virtual consultations. Similar to this, the Philippine Statistics Authority (2022) documented a significant increase in pet ownership, with 67% of households owning at least one pet. This trend underscores the demand for comprehensive veterinary solutions that cater to a growing population of pet owners. Moreover, the Animal Kingdom Foundation (2021) reported that public awareness of preventive pet healthcare remains low, advocating for platforms that provide vaccination reminders and health management tools. The Department of Information and Communications Technology (DICT) emphasized the role of digital platforı

through its Digital Cities 2025 Initiative (2020), aimed at enhancing service delivery across various sectors, including veterinary care.

This supports PetEat's inclusion of features like electronic medical records (EMRs) to streamline operations and foster connections among pet owners. Additionally, the Veterinary Science Innovations Journal (2022) highlighted the early adoption of NFC tracking technology in the Philippines, showcasing its potential to address pet safety concerns and improve recovery rates for lost pets.

2.2 Foreign Literature

Telemedicine has become a significant part of veterinary medicine globally. The Innovative Veterinary Care Journal (2023) highlighted that telemedicine adoption accelerated during the COVID-19 pandemic, transitioning from limited phone consultations to virtual video consultations. Telemedicine appeals to pet owners for its convenience, particularly for those with busy schedules or anxious pets. However, challenges such as technological limitations and the inability to perform physical examinations persist. Cosgrove (2025) emphasized the growing issue of pet loss, with approximately 10 million cats and dogs stolen or lost annually in the United States. NFC tags have proven effective in improving pet recovery rates, a feature that PetEat plans to incorporate. According to Douzet (2024), technology enablement will help bridge the veterinary supply and demand gap, with telemedicine and at-home diagnostic tools such as tracking collars and smart litter boxes offering efficient alternatives. Gibson (2024) discussed the problem of animal abandonment, exacerbated by economic challenges. Digital platforms that connect pet owners with shelters can

mitigate this issue by simplifying adoption processes and offering resources for struggling owners. Singh et al. (2021) highlighted the efficiency of online appointment scheduling systems in healthcare. Automated scheduling reduced administrative burdens and minimized appointment errors, directly informing PetEat's appointment scheduling feature.

2.3 Local Studies

Exponential expansion of the pet care market in the Philippines triggered innovative means to solve the existing problems associated with pet ownership and veterinary clinics. Many local and international surveys from 2019-2023 provide a basis for this research as they show changes and development in integrated pet care systems. The Philippine E-Journals (2023) published a feasibility study on the Paws and Fur Park Hotel, which proposed an integrated veterinary care center offering boarding and grooming services. This aligns with PetEat's goal to centralize pet care services. A study titled Designing Pet Care: An IoT Approach for Integrated and Empowered Pet Care at a Distance (2019) explored the application of IoT for remote pet care. The system emphasized real-time support when owners are not physically present, similar to PetEat's NFC and remote monitoring features. The research paper Implementation of a Pet Care Management System (2023) highlighted the importance of comprehensive web applications for pet care, including functionalities like medical recordkeeping and appointment scheduling—key elements of PetEat. The Integrated Digital Platform for Pet Care Services: The Paw Labs (2021) study presented an innovative solution for bridging the gap between pet care providers and owners,

supporting the necessity for platforms that harmonize online and offline services. Lastly, the IoT-Based Pet Care System (2020) illustrated how automation and cloud integration enhance pet care solutions, advocating for scalable systems such as PetEat in urban settings.

2.4 Foreign Studies

A study by Guercio et al. (2020) explored the adoption of telemedicine in veterinary practice, particularly in Europe and North America. The study highlighted how telemedicine has improved accessibility for pet owners, especially in rural areas, allowing them to consult with veterinarians remotely. Through mixed-method research, they found that telemedicine not only reduced barriers to care but also increased client satisfaction. Their findings concluded that integrating telemedicine into veterinary practices could reduce clinic workloads and offer pet owners a convenient alternative for non-emergency consultations. This aligns with the PetEat system's aim of providing telemedicine services to enhance pet care accessibility. A research study by Park et al. (2019) examined the effectiveness of near-field communication (NFC) technology for animal identification and tracking. Conducted in South Korea, the study analyzed how NFC tags could improve the identification process for pets and livestock compared to traditional methods like microchips. Their experimental results demonstrated that NFC technology significantly reduced the time required to locate and identify lost animals. These findings emphasize the practicality of NFC in the PetEat system, particularly for pet tracking and recovery features. A study by Singh et al. (2021) explored the efficiency of online appointment scheduling systems in human healthcare and its potential application in veterinary services. Conducted in Australia, the study found that

automated scheduling reduced administrative burdens, minimized appointment errors, and improved overall client satisfaction. Their findings directly inform the development of the PetEat system's scheduling functionality, which aims to simplify the booking process for veterinary clinics and pet owners. A study by Chen et al. (2020) analyzed the factors influencing the adoption of mobile applications in veterinary services in China. The study identified ease of use, perceived usefulness, and data security as the most significant factors driving user acceptance. The researchers recommended incorporating intuitive navigation and secure payment systems to boost adoption rates. These insights directly inform the PetEat system's emphasis on user-friendly design and secure transaction processing. A study conducted by Ortiz et al. (2019) in Spain evaluated the impact of digital platforms in managing animal shelters. The researchers found that implementing digital record systems for pet profiles, medical histories, and adoption processes significantly reduced administrative errors and improved adoption rates. Their findings validate the PetEat system's potential to streamline pet records and enhance user interactions, whether for core clinics or pet shelters. A study by Brown et al. (2020) investigated the use of GPS and NFC technologies in pet tracking within urban areas of the United States. The research found that NFC tags provided a cost-effective and efficient alternative to GPS for identifying lost pets, especially in densely populated areas. Their study supports PetEat's decision to incorporate NFC tracking as a feature.

2.5 Synthesis

The literature highlights the transformative potential of digital technologies in addressing gaps in pet healthcare. Local studies stress the need for integrated solutions due to growing pet ownership and limited veterinary services, while international research emphasizes the efficiency and accessibility of telemedicine and IoT tools. PetEat's development aligns with these findings, demonstrating the value of technology-driven approaches to improve veterinary care and operational efficiency.

Chapter 3

METHODOLOGY

This chapter outlines the research methods and processes utilized in the development of the PetEat platform. It presents a structured approach to data collection, system design, and feasibility analysis, ensuring the effectiveness and efficiency of the system in addressing the needs of veterinary clinics and pet owners. This chapter also provides a comprehensive examination of the system's architecture and the feasibility of its implementation.

3.1 Requirements Analysis

The requirements analysis phase evaluates the feasibility of developing and implementing the PetEat platform by assessing its operational, technical, economic, and schedule feasibility. These factors ensure that the system effectively meets user expectations and remains practical throughout its lifecycle.

Operational Feasibility

Operational feasibility assesses PetEat's practical application in veterinary clinics and pet care settings. The system is designed for ease of use, facilitating smooth interactions between pet owners and veterinary professionals. Features like appointment scheduling, NFC reporting, and EMR management enhance efficiency. Usability testing confirms that PetEat is accessible and requires minimal training for adoption.

Technical Feasibility

The technical feasibility determines whether the technology used in PetEat supports its functionalities and scalability. The system integrates web and mobile applications, employing the following tools such as:

Table 3.1

Software Sequence of Project Development

Development	For Web	For Mobile
Frontend	React.js for web	Flutterflow for mobile
Backend	Node.js with Express	
Database	MySQL	Firebase

Table 3.2

Minimum and Recommended Hardware Requirements for Development

Requirement	Minimum	Recommended
Processor	Ryzen 5 5600G	Ryzen 5600X
RAM	16GB	32GB
Storage	50GB of Free Space	100GB of Free Space
Operating System	Windows 10	Windows 11
Graphics Card	AMD Integrated Graphics	Nvidia RTX 3060Ti

Table 3.1 presents the development tools used for web and mobile applications. It categorizes them into frontend, backend, and database technologies. For web applications, React.js is used for the frontend, Node.js with Express for the backend, and MySQL for the database. For mobile applications, Flutterflow is used for the frontend, while Firebase serves as the database solution. Table 3.2 outlines the minimum and recommended PC requirements for development. It compares hardware and software specifications, including processor, RAM, storage, operating system, and graphics card. The minimum setup includes a Ryzen 5 5600G processor, 16GB RAM, 50GB of free storage, Windows 10, and AMD Integrated Graphics. The recommended setup suggests a Ryzen 5600X processor, 32GB RAM, 100GB of free storage, Windows 11, and an Nvidia RTX 3060Ti graphics card for better performance.

Table 3.3

Minimum and Recommended Mobile Requirements for Deployment

Requirement Type	Minimum	Recommended
Processor	Quad - Core Processor	Octa - Core Processor
RAM	4GB	8GB
Storage	3GB of Free Space	5GB of Free Space
Operating System	Android 8.0	Android 15
Network	4G / LTE	5G
NFC	Enabled	Enabled

Table 3.3 outlines the minimum and recommended mobile requirements for deployment. The minimum setup includes a quad-core processor, 4GB RAM, 3GB free storage, Android 8.0, 4G/LTE support, and NFC capability. The recommended setup

features an octa-core processor, 8GB RAM, 5GB free storage, Android 15, 5G support, and NFC capability for optimal performance.

Table 3.4

Minimum and Recommended Hardware Requirements for Development

Requirement Type	Minimum	Recommended
Software	Chrome: A later version of Chrome 40 series Opera: Opera 27 Firefox: Firefox 43 Edge: Microsoft Edge version 25.10586	Chrome: 130 or newer Opera: 80 or newer Firefox: 130 or newer Edge: 130 or newer
Hardware	Windows 10,8GB Ram, CPU 1Ghz (Intel Celeron or Greater)	Windows 11, 16GB Ram, CPU 4.0Ghz (AMD Ryzen 5 2600 or Greater)

The table outlines the minimum and recommended PC requirements for running the website. The minimum requirements for the web browser are Chrome: A later version of Chrome 40 series, Opera: Opera 27, Firefox: Firefox 43, Edge: Microsoft Edge version 25.10586, while the recommended specifies Chrome: 130+, Opera: 80+, Firefox: 130+, Edge: 130+, Additionally this also specifies the necessary operating system, RAM, and processor capabilities. The minimum requirements include Windows 10, 8GB RAM, and a 1GHz processor (Intel Celeron or greater). The recommended setup suggests Windows 11, 16GB RAM, and a 4.0GHz processor (AMD Ryzen 5 2600 or greater) for better performance and smoother browsing experience.

Economic Feasibility

A cost-benefit analysis ensures that PetEat is financially sustainable. The system minimizes costs by utilizing cloud-based services like Firebase and Amazon Web Services (AWS). The expected benefits include improved veterinary services, efficient management of pet medical records, and enhanced user engagement.

Table 3.5

Cost-benefit Analysis Table

Category	Cost	Benefits
Development	Software, Cloud services, API integration	Scalable system, Improved User Experience
Operations	Customer Support	Reliable Platform, Continuous improvement
Regulations	Compliance with data privacy laws	Trust and Credibility

The cost-benefit analysis table evaluates the financial sustainability of PetEat by comparing costs and expected benefits across key categories. It highlights the use of cloud services like Firebase and AWS to minimize expenses while ensuring scalability, user experience, and reliability.

Table 3.6

Developmental Cost Table

Requirement Type	Minimum	Recommended
Hardware		
PC Requirements for Development	₱28,000.00	₱45,000.00
Software		
Front End (React.js for Web, Flutterflow for Mobile)	₱0.00	₱2,500.00 (Subscription Plan)
Back end (Node.js with Express)	₱0.00	₱0.00
Database (MySQL and Firebase)	₱0.00	₱500.00 (Subscription for more storage)
Operational Expenses		
Electric Bill	₱2,500.00	₱2,500.00
Transportation	₱1,500.00	₱1,500.00
Printing of Documents	₱500.00	₱500.00
Total Expenses:	**₱32,500.00**	**₱52,500.00**

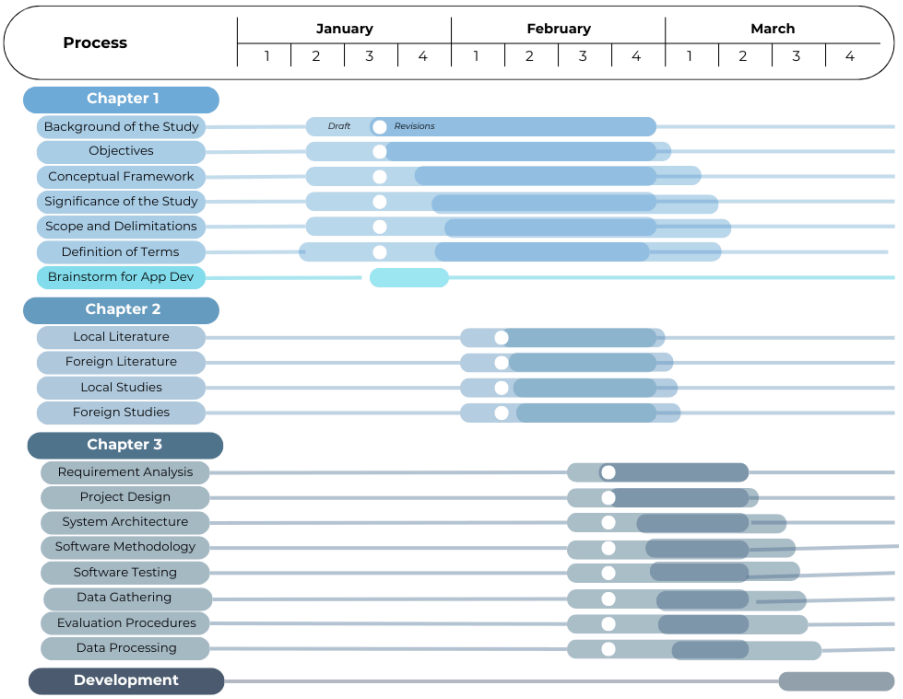
Table 3.4 outlines the development costs for the PetEat platform. Many technologies, such as React.js, Node.js with Express, and MySQL, are free, reducing initial costs. However, Flutterflow, a no-code mobile development tool, starts at ₱2,500, and Firebase may cost at least ₱500, depending on usage. The total estimated cost ranges from ₱0 to ₱3,000, mainly influenced by the need for premium services.

Schedule Feasibility

The project timeline follows an Agile development model, ensuring iterative progress through defined sprints. A structured development schedule guarantees timely completion while accommodating necessary improvements based on user feedback.

Figure 3.1

Gantt Chart



The Gantt chart illustrated in Figure 3.1 outlines the structured timeline for the project. The researchers progress through iterative refinements of Chapter 1, including revising the title, tackling the chapter's content, and finalizing its structure. Key brainstorming activities for the app take place mid-January, ensuring a well-planned development process. Moving into February, the focus shifts towards Chapter 2 and

Chapter 3, where drafting, revisions, and manuscript finalization occur. By March, efforts transitioned towards the development of the web and mobile applications, indicating a shift from documentation to implementation.

3.2 Project Design

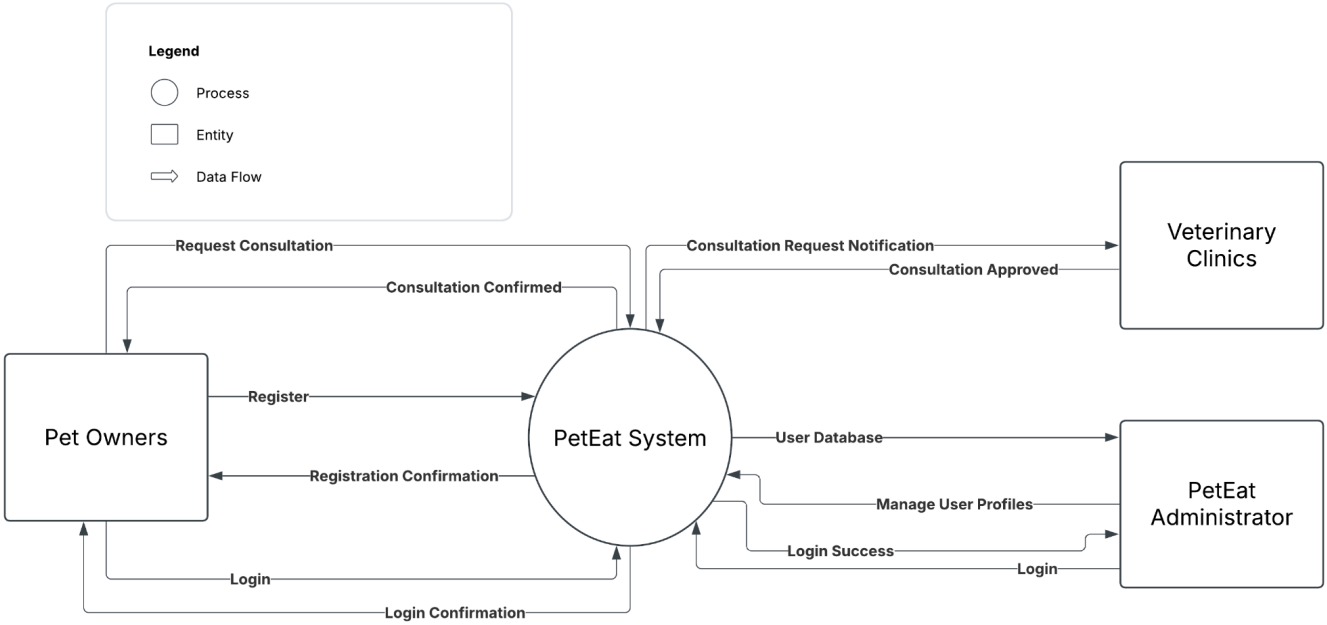
The system design provides a comprehensive framework for the functionalities and structure of the PetEat platform. It ensures that the application meets its objectives while maintaining a seamless user experience. The following diagrams illustrate the system's architecture and interactions.

Context Diagram

The context diagram provides an overview of how external entities interact with the PetEat system, including pet owners, veterinary clinics and suppliers. As illustrated in Figure 3.2, the context diagram provides a high-level overview of the PetEat System and its interactions with key users: PetEat Admin, Pet Owners, and Veterinary Clinics. Pet Owners can register, log in, and input their pet's information into the system. Veterinary Clinics interact with the system by managing medical appointments and conducting online consultations. Meanwhile, the PetEat Admin oversees user profiles and handles profile management. This diagram highlights the structured data flow within the system, ensuring efficient pet healthcare management, user engagement, and appointment coordination.

Figure 3.2

Context Diagram



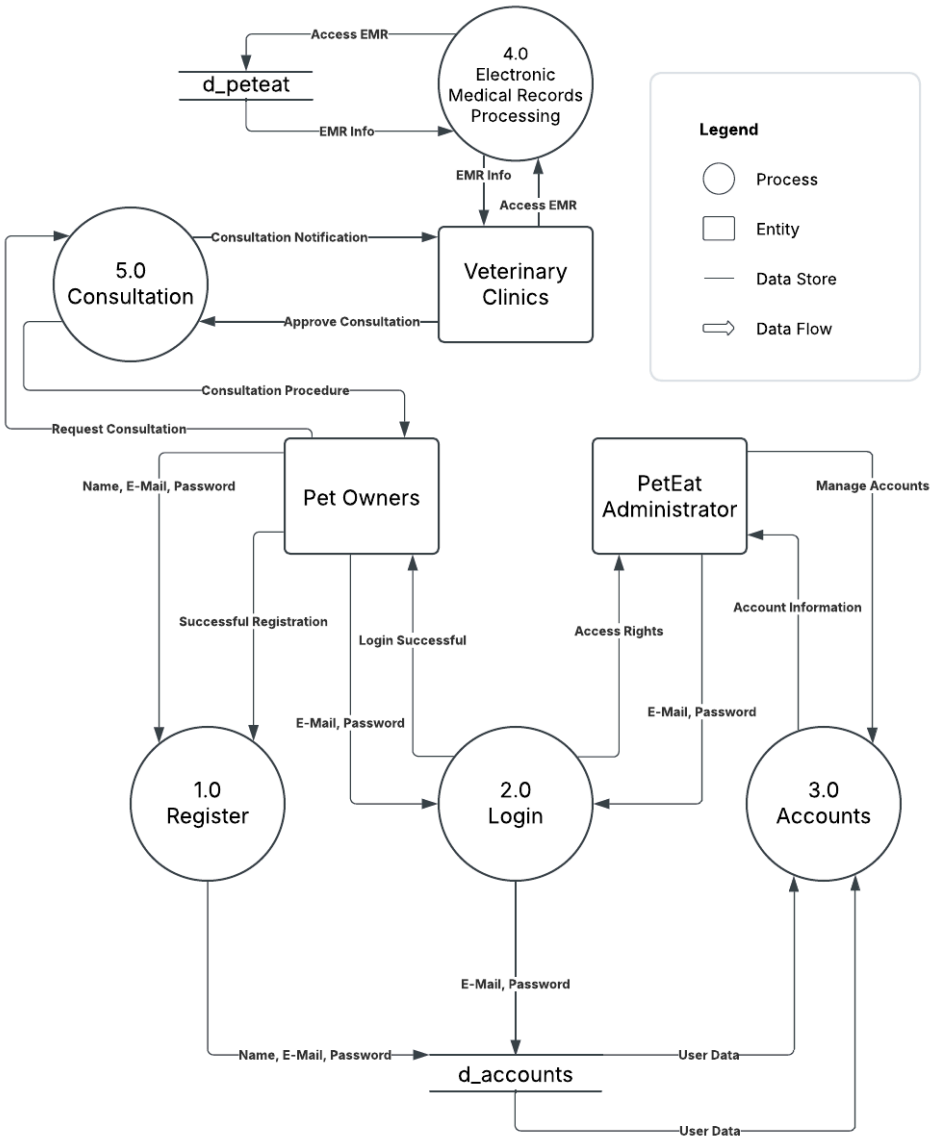
Level 1 DFD Diagram

The Level 1 DFD presents a detailed breakdown of a system's processes and how they interact with users and data storage, including vet clinic selection and interactions The PetEat system, as depicted in the Level 1 Data Flow Diagram DeMarco & Yourdon Notation (Figure 3.3), outlines a comprehensive platform designed to facilitate pet recovery, and veterinary consultation. The system initiates with a robust user registration process, enabling individuals to create accounts by providing essential details such as name, email, and password, subsequently stored securely within the system's data stores. Upon successful registration, users can authenticate themselves through a login process, granting them access rights to the system's functionalities. Registered users can leverage the system for veterinary

consultations, requesting video sessions with veterinary clinics and accessing stored medical records within the PetEat Cloud Storage. Furthermore, the PetEat Administrator plays a crucial role in managing user accounts and moderating community posts, ensuring the platform's integrity. Overall, this DFD illustrates a multifaceted system that integrates various functionalities to support pet owners and enhance pet welfare.

Figure 3.3

Level 1 DFD Diagram (DeMarco & Yourdon Notation)

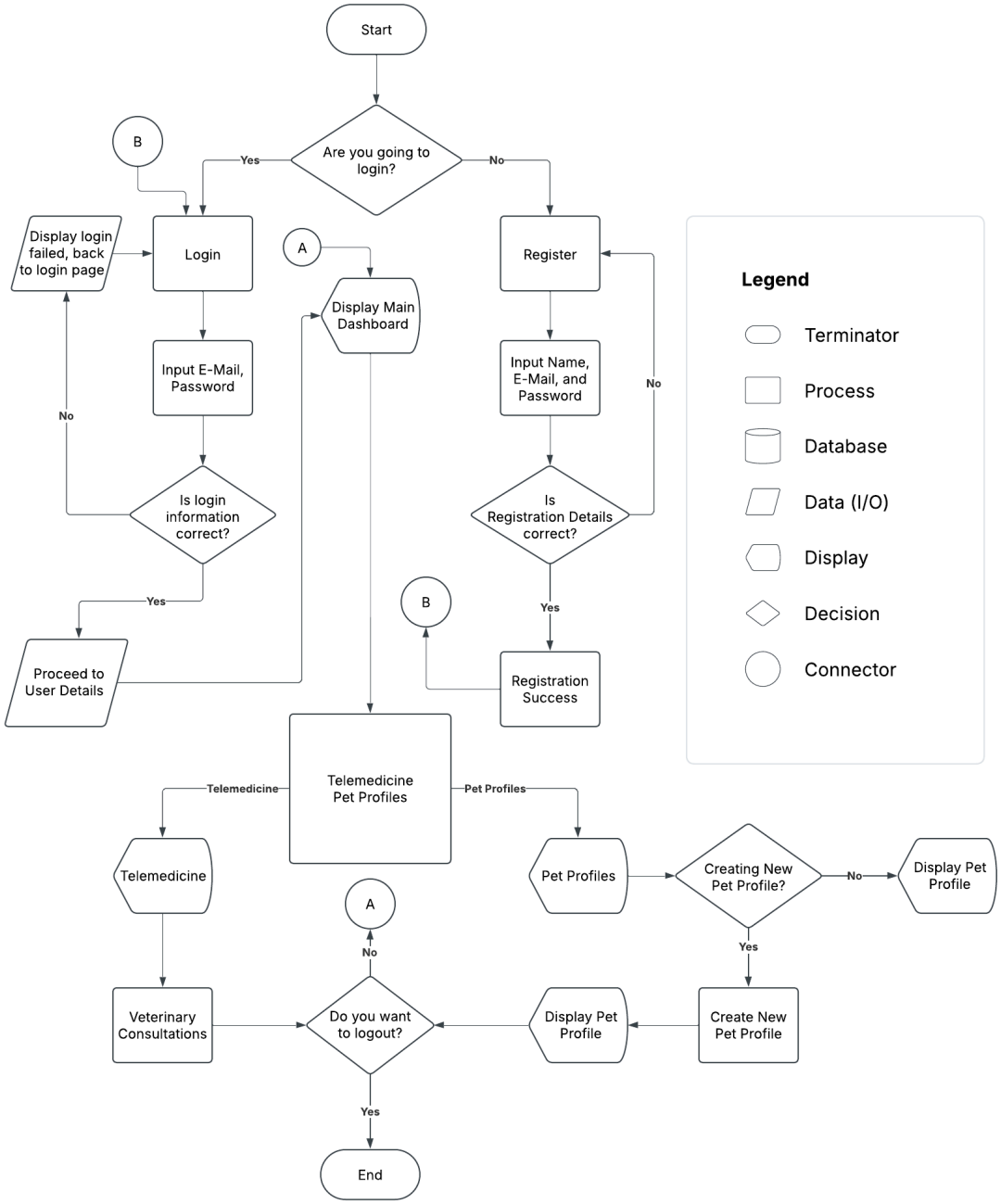


System Flowchart

The system flowchart outlines the logical sequence of operations, including requesting a video consultation. The system flowchart outlines the logical sequence of operations, including requesting a video consultation. The system flowchart illustrated in Figure 3.4 outlines the user flow and system logic for a pet-focused application, likely incorporating telemedicine and pet profile management. The process begins with the user choosing to either login or Register. If they choose to login, they are prompted to enter their email and password. The system then checks if the login information is correct. If incorrect, the user is redirected back to the login page with an error message. If correct, the user proceeds to the Main Dashboard and User/Admin Details. For Registration, the user inputs their Name, E-Mail, and Password. The system checks for successful registration. If successful, the user receives a confirmation and proceeds to the main dashboard. If unsuccessful, they are prompted to re-enter their details. The main dashboard provides access to several key features: telemedicine, and pet profiles. Telemedicine allows users to request consultations with a vet clinic. Pet profiles let users create and manage profiles for their pets. Users can also logout from any point within the application. The flowchart demonstrates a clear pathway for users to access and utilize the application's features, covering both login/registration and the core functionalities of the platform.

Figure 3.4

System Flowchart

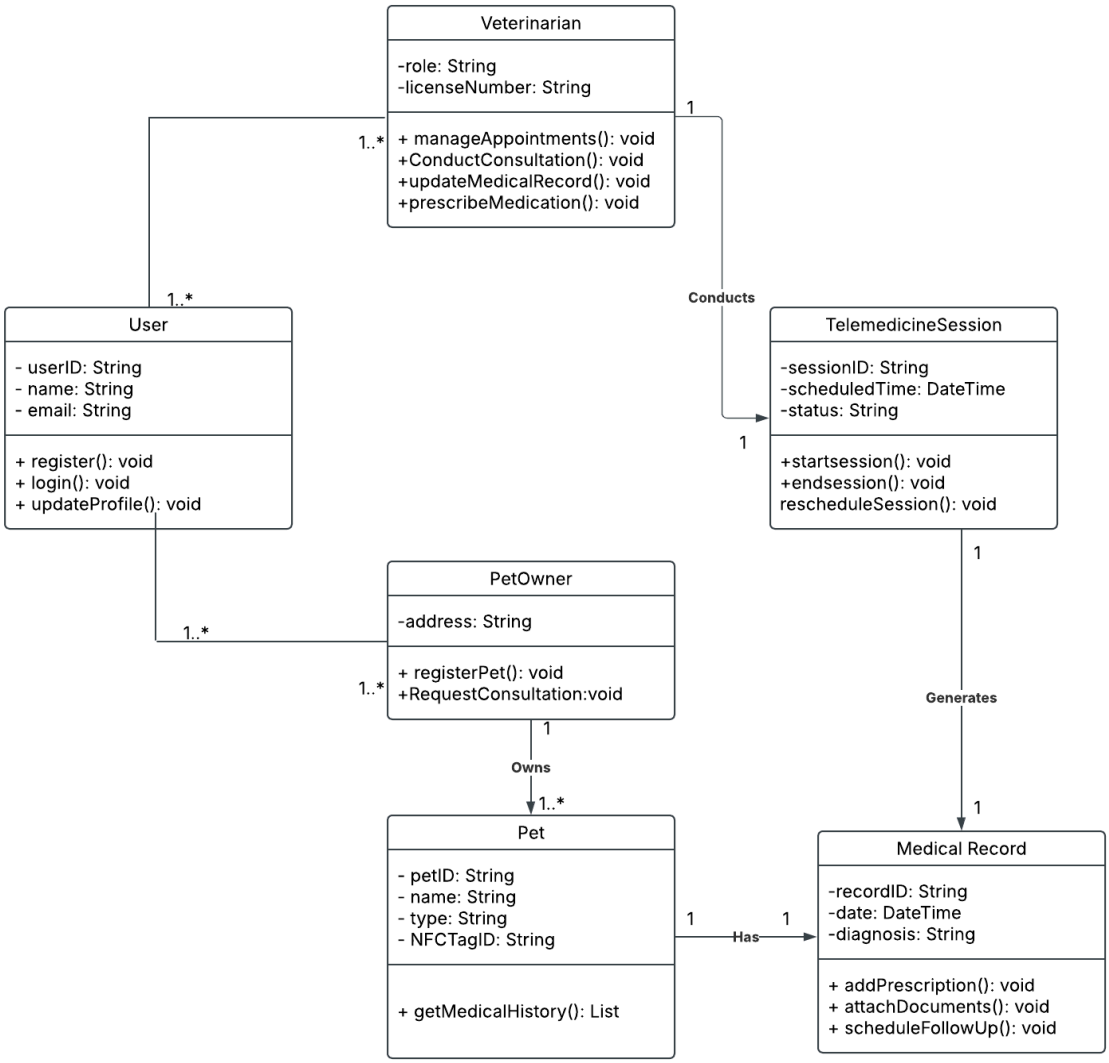


UML Class Diagram

A standardized modeling language used to design and visualize software systems to represent different aspects of a system's structure and behavior. The UML Class Diagram illustrated in Figure 3.5 illustrates the structure of a pet management system by defining various entities and their relationships. At the core of the system is the User class, which includes attributes such as userID, name, and email, along with methods for registration, login, and profile updates. Extending from the User class, the PetOwner class incorporates additional attributes, such as address, and provides functionalities for registering pets and requesting consultations. The Veterinarian class, which includes attributes like role and licenseNumber, plays a crucial role in managing appointments, conducting consultations, updating medical records, and prescribing medications. The Pet class holds essential pet details, including petID, name, type, and NFCTagID, with a method to retrieve medical history. Pet health records are managed by the MedicalRecord class, which maintains attributes such as recordID, date, and diagnosis, offering functionalities to add prescriptions, schedule follow-ups, and attach documents. The TelemedicineSession class facilitates remote veterinary consultations, containing attributes like sessionID, scheduledTime, and status, along with methods to start, end, and reschedule sessions.

Figure 3.5

UML Class Diagram

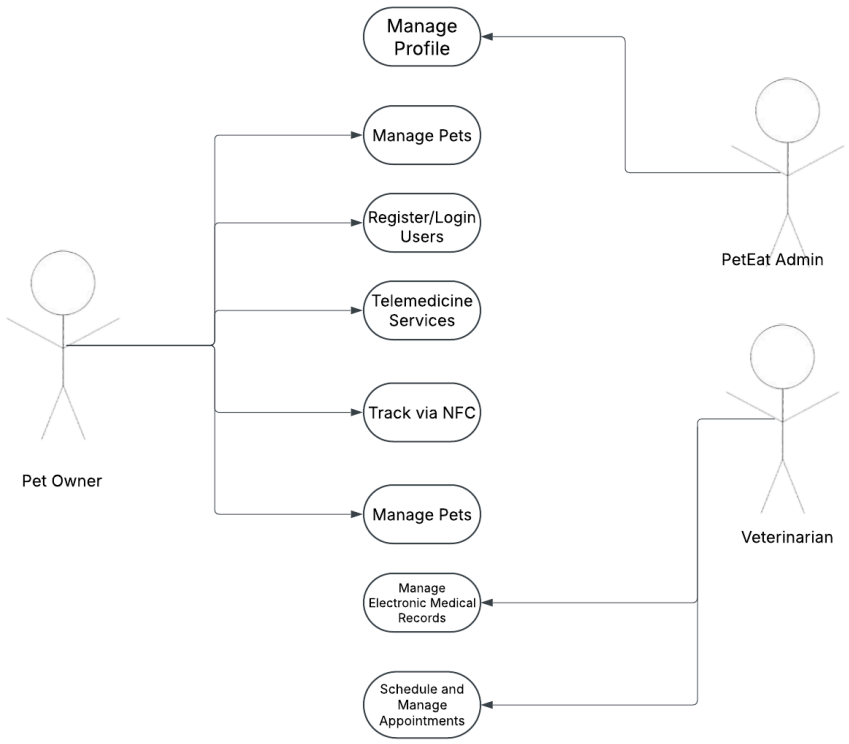


Use Case Diagram

This represents the functional requirements of a system by showing how users (actors) interact with the system's features (use cases). It helps visualize what the system does and who can perform each action. The Use Case Diagram illustrated in Figure 3.6 illustrates the interactions between different user roles and the system's functionalities in a pet management system. It identifies three primary actors: Pet/Est Admin, Veterinarian, and Pet Owner, each associated with specific system functions. The PetEat Admin is responsible for high-level management tasks such as Profile Management. The Veterinarian interacts with the system through features like Telemedicine, ensuring efficient medical services and record-keeping. The Pet Owner engages with the system by Managing Pets, Logging in/Registering, and utilizing NFC Tracking for pet identification

Figure 3.6

Use Case Diagram

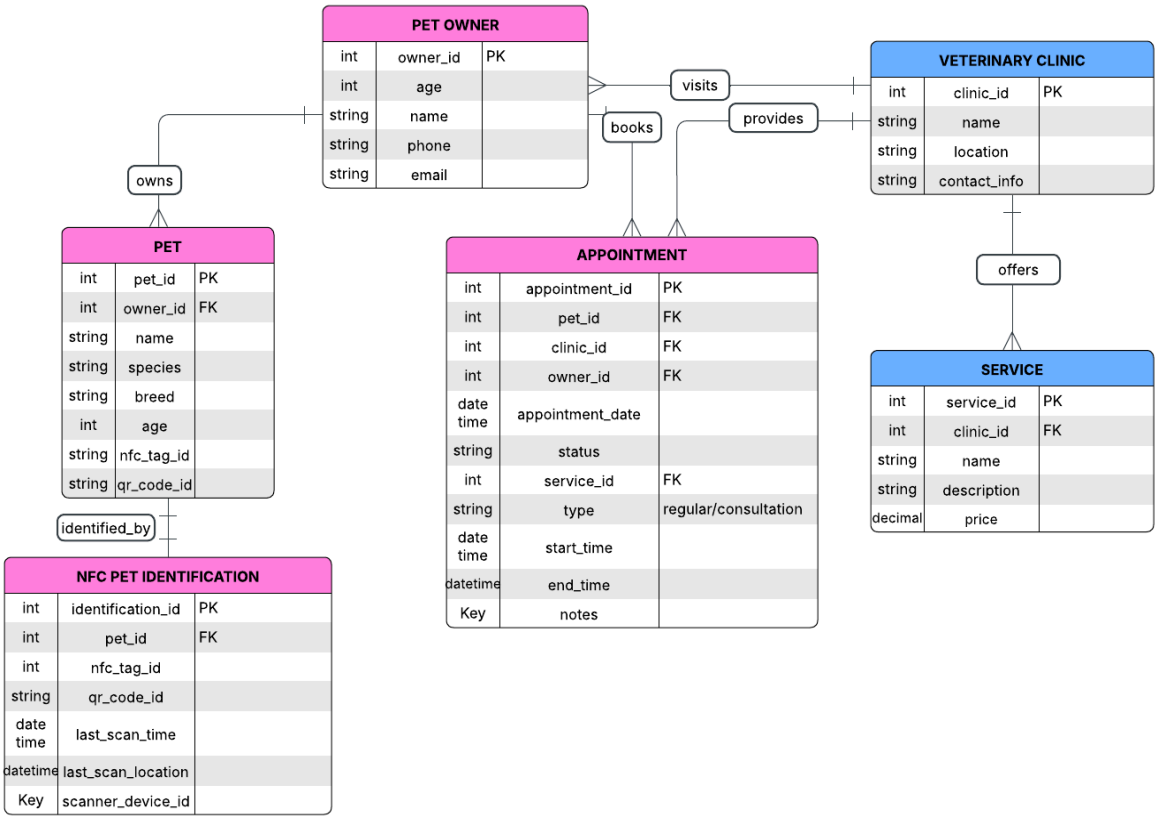


Entity Relationship Diagram

An Entity Relationship Diagram (ERD) is a visual representation of a database structure that illustrates how entities (objects or concepts) relate to one another within a system**.** The Entity Relationship Diagram illustrated in Figure 3.7 is designed to enhance veterinary clinic operations and advance management of pet care. Basically, the system is the platform in which pet owners can register the pets and reserve routine appointments or consultation sessions in veterinary clinics. The appointments either are routine visitations or consultancy sessions, whose respective time and comments are inputted into the same system. NFC technology is utilized for pet identification, enabling quick owner retrieval, while clinics can efficiently manage their services.

Figure 3.7

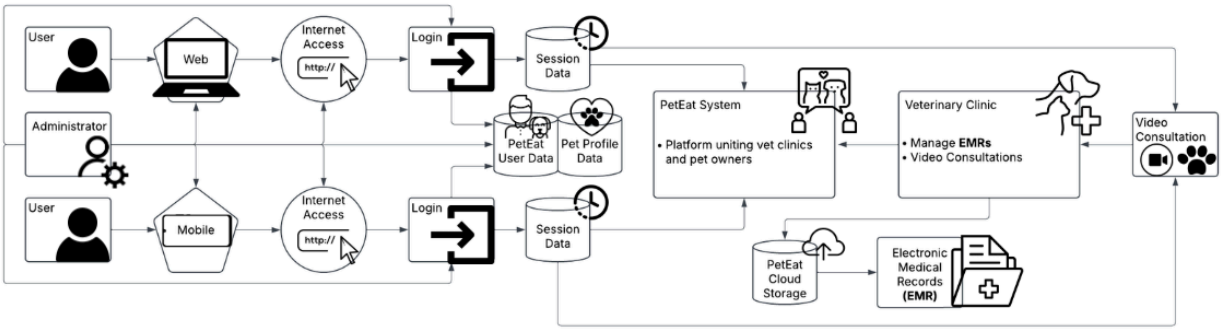
Entity Relationship Diagram



3.3 System Architecture

The PetEat platform's research architecture follows a modular design to enhance flexibility and maintainability. The user interface layer serves as the primary interaction point for pet owners and clinic staff, providing functionalities such as record management, telemedicine scheduling, notifications, and user authentication. The application layer handles core business logic, including the storage and retrieval of pet health records from the data layer, while also managing APIs for seamless operation. The integration layer facilitates communication with external systems, such as telemedicine platforms, pet health monitoring devices, payment gateways, and veterinary databases, ensuring interoperability and extended functionality.

Figure 3.8



System Architecture Diagram

The System Architecture Diagram (Figure 3.8) outlines the interconnected components of the PetEat platform, a web and mobile application designed for pet owners. The diagram illustrates the flow of information and interactions between various user roles and system functionalities. At the core, the User interacts with the

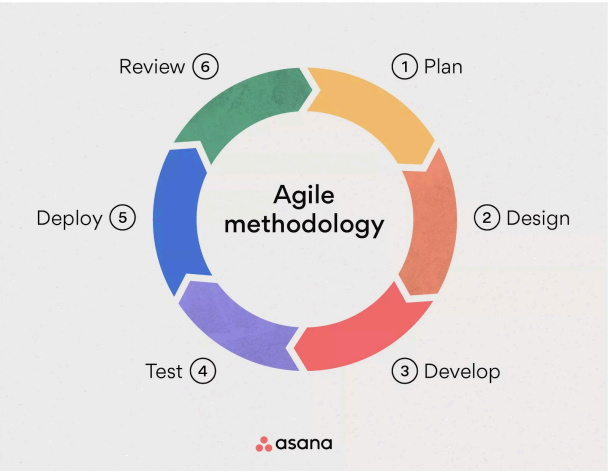
system through registration, pet management, and video consultation requests. This user activity is facilitated by Internet Access and stored within the PetEat Cloud Storage, which also houses Electronic Medical Records (EMR). The Administrator role oversees user management, content moderation, access control, and system monitoring, ensuring the platform's integrity. Additionally, Veterinary Clinics manage EMRs, conduct video consultations, and provide expert advice to the community. This architecture effectively integrates pet management, and veterinary services, creating a platform for the pet owners and veterinary clinic.

3.4 Software Development Methodology

This project will employ the Agile development methodology due to its flexibility and rapid iteration across all stages of development. Agile enables continuous feedback and refinement through iterative sprints, ensuring that the system evolves based on real-world needs.

Figure 3.9

Agile Methodology Model (Anito, 2024)



Planning

The process begins with defining the project scope, objectives, and key requirements based on insights from veterinarians, pet owners, and clinic staff. During this phase, the researchers outline core system features such as telemedicine, EMR (Electronic Medical Records) and inventory management while establishing a project roadmap and user stories to guide development.

Design

In these phases, the researchers create a system architecture diagram to provide a structured blueprint for development. This phase includes the preparation of Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), UML diagrams, and Use Case Diagrams to define system components clearly. Additionally, the database structure and API endpoints are planned to ensure a streamlined development process.

Development

The system is then implemented in iterative sprints, allowing for incremental development and testing of key functionalities. The researcher develops features such as: Appointment scheduling, Electronic Medical Records (EMR), NFC-based pet tracking identification. The technologies used are: React.js (Web), Flutter (Mobile), and Node.js (Backend). Each feature undergoes individual testing before integration to maintain software quality. user-driven manner, ensuring that it effectively meets the needs of veterinary clinics and pet owners while adapting to evolving requirements.

Testing

Comprehensive testing ensures that the system functions correctly and meets user expectations. This process includes various testing phases, such as unit testing, which validates individual modules, and integration testing, which ensures seamless interaction between different components. Additionally, performance, security, and usability testing are conducted to assess the system's efficiency, safety, and user-friendliness. Finally, User Acceptance Testing (UAT) is carried out with veterinarians and pet owners to gather real-world feedback, ensuring the system meets practical needs and expectations.

Deployment

Once validated, the system is deployed on cloud servers for initial use. Pilot testing is conducted in selected veterinary clinics, allowing real users to provide feedback and identify areas for improvement before full-scale implementation.

Maintenance and Feedback

The researcher continuously monitors system performance, resolves issues, and integrates user feedback to enhance features. Security updates and scalability improvements are also implemented to ensure long-term sustainability and system efficiency. By following the Agile methodology, the PetEat platform is developed in a flexible, iterative, and user-driven manner, ensuring it effectively meets the needs of veterinary clinics and pet owners while adapting to evolving requirements.

3.5 Software Testing

To ensure the proper functioning of the PetEat platform, a comprehensive testing approach will be implemented, covering both automated and manual testing methods. Unit testing will be conducted by developers and QA testers using automated tools to identify and rectify errors in individual features such as appointment scheduling, electronic medical record (EMR) retrieval, and NFC tracking. Once individual components are validated, integration testing will be performed to assess how different modules, such as telemedicine, inventory management, and the payment processor, interact. Developers and testers will observe data flow and resolve compatibility issues between these modules to ensure seamless operation.

System testing will follow, focusing on evaluating the platform's overall performance, security, and usability in real-world conditions. This phase will include stress testing to assess system behavior under high loads and security testing to identify vulnerabilities. Manual testing will also be incorporated throughout different phases, where QA testers will manually execute test cases to assess user interface behavior, system workflows, and overall user experience.

Additionally, both white box and black box testing methodologies will be applied. White box testing will focus on internal structures and logic, ensuring that the source code executes correctly through techniques such as statement coverage, branch coverage, and path coverage. On the other hand, black box testing will be used to evaluate the system’s functionality without requiring knowledge of the internal code. This approach will involve functional testing to validate features like appointment

scheduling and telemedicine, usability testing to assess user-friendliness, compatibility testing across different devices and browsers, regression testing to ensure that new updates do not introduce errors, and security testing to identify potential threats.

Finally, User Acceptance Testing (UAT) will be conducted with veterinarians, clinic personnel, and pet owners to assess the system's usability in real-world scenarios. Key features such as pet medical records, telemedicine, and vaccination reminders will be evaluated, and feedback will be collected through surveys. The insights gathered from UAT will be used to refine and enhance the platform, ensuring it meets user needs and expectations.

3.6 Data Gathering and Techniques

In this research study, quantitative data collection methods were used. The researchers initially collected all information and facts from articles and reviews to support the study, and the material needed in the study through interviews. Then, the researchers prepared an instrument or survey to be given to the respondents using Google Forms for better and faster accessibility, as we are using modern technology today. The data for this research were collected using a survey questionnaire modified and adopted from the International Standard Organization (ISO/IEC 25010:2011) and formed by the researchers.

Population and Sample

The population of this study consists of individuals involved in pet care and veterinary services within Muntinlupa. To ensure a well-rounded perspective, the study will include two groups of respondents: pet owners and professionals in the veterinary and IT fields. A total of 40 participants will be selected through purposive sampling. This includes 20 pet owners who represent the primary users of the system, 10 veterinarians who provide insights into clinic operations and pet healthcare, and 10 IT experts who evaluate the system's technical feasibility. The selection is limited to Muntinlupa to maintain relevance and practicality in system implementation.

Table 3.7

Sample Table

Respondents	Distribution
Non-Experts (Pet Owners)	20
Non-Experts (Veterinarians)	10
Experts (IT Experts/Professionals)	10
Total	**40**

To investigate the respondent's perceptions about PetEat: A Web & Mobile Platform for Veterinary Clinics and Pet Lovers with NFC Tracking, Telemedicine, and Cloud Based EMR, the questionnaire was administered to pet owners, veterinarians and IT experts. during the 3rd Term of the school year. The questionnaire will focus on the respondents' assessment on the PetEat: A Web & Mobile Platform for Veterinary

Clinics and Pet Lovers with NFC Tracking, Telemedicine, and Cloud Based EMR . Before doing the survey, the researcher will provide a letter to ask permission to conduct the study. A letter for the respondents will provide for their approval. They were not forced to answer the questions included in the survey questionnaire. About 10-15 minutes was the allotted time to finish the distribution of the questionnaire. After the given time, the researchers will collect and organize the data gathered from the participants and a deep interpretation of the results is included in the study. In the questionnaire, the Likert scale will be used to determine the user satisfaction and areas for improvement. The researchers assured confidentiality of their survey sheets since the identities are not important. The researchers also understood that people's consciousness may also affect their honesty and effectiveness in answering the survey. Thus, the researchers gave people the option of being anonymous. The participants were given time to respond and then the researchers collected the questionnaire on the next day.

3.7 Evaluation Procedures

To ensure an efficient evaluation of PetEat, various methods will be conducted to assess the system's functionality, usability, performance, and overall effectiveness in which the researcher used an adopted model from ISO 2502011 in addressing the needs of veterinary clinics and pet owners (see the ISO 250:2011) and updates the characteristic. This evaluation process plays a vital role in helping identify potential issues, test system optimization, and guarantee that the platform meets user expectations prior to releasing the platform into operation. The system keeps its

performance standard with accurate information delivery and continues providing service to its intended user group. The process helps reduce technical problems and results in decreased maintenance expenses, together with better application reliability.

The evaluation consists of usability testing together, performance testing, security assessment, and user acceptance testing (UAT). Below are the ff Quality Attributes of ISO 250:2011 Software Quality Evaluation Indicators

Quality Attributes

Usability. In usability testing, participants will be asked to complete specific tasks, such as scheduling appointments and accessing pets' medical records, while being observed. This process will be performed through surveys and interviews to identify areas for improvement.

Functional Suitability. In functional suitability, this ensures that the system provides all necessary features to meet user needs. Testing will focus on verifying the correct operation of core functionalities, including telemedicine consultations, NFC tracking for lost pets, electronic medical record (EMR) management and inventory management.

Performance Efficiency. In performance efficiency, this measures the system's response time, resource utilization, and scalability. Key features to be tested include the response time of appointment booking, server performance under high traffic, database query execution speed, and the quality of video calls during telemedicine consultations.

Compatibility. In compatibility, it ensures the system operates effectively across different devices, operating systems, and third-party integrations. The web versions will be tested on various browsers such as Chrome, Firefox, Safari, and Edge to validate

cross-browser functionality. The mobile versions will be tested within the minimum and recommended requirements to ensure a wide range of compatibility on mobile smartphones.

Reliability. In reliability, this assesses the software's stability and ability to function under specified conditions without failure. System uptime, failure rate, and recovery time will be monitored to ensure continuous and uninterrupted operation. The key features to be tested include functionality for medical records and consultations and seamless data synchronization between web and mobile platforms to ensure uninterrupted service.

Security. In security, this ensures the protection of user data from unauthorized access, breaches, and cyber threats. Features to be tested include user authentication with role-based access control, data security for medical records, prevention of unauthorized modifications to pet profiles, and secure payment processing.

Maintainability. In maintainability, this measures how easily the software can be modified, updated, or fixed. The system's architecture will be evaluated for code complexity, modularity, and ease of updates to ensure long-term adaptability.

Portability. In portability, this assesses how easily the software can be transferred or adapted to different environments. Testing will include deployment on various platforms to ensure flexibility and ease of migration.

User Acceptance Testing (UAT). InUser Acceptance Testing (UAT), this verifies that the system meets end-user requirements and expectations. Telemedicine and Cloud Based EMR will be tested for practicality and ease of use, while feedback will be collected to refine the platform based on user needs. This testing phase will

ensure that the system aligns with business requirements and delivers a positive experience for veterinary clinics and pet owners.functionality, usability, and overall satisfaction before final deployment.

3.8 Data Processing and Statistical Treatment of Data

The data gathered from testing, surveys, and interviews will be processed and analyzed from collection to organization as well as using descriptive statistics and distribution in order to attain insights for system improvement.

Data Processing

Responses from surveys, unstructured interviews, and user testing will be gathered through data collection, followed by removing or correcting inconsistencies and incomplete data entries by data cleaning. And finally, arranging the information in tables, charts, and graphs to facilitate analysis through data organization.

Statistical Treatment of Data

The frequency and percentage of responses will be calculated to assess trends from user experience and feedback with frequency distribution, while mean, median, and standard deviation will be used to analyze survey responses regarding evaluation procedures (Usability Testing Together, Performance Testing, Security Assessment, System Accuracy and Reliability, and User Acceptance Testing (UAT) to attain quantitative data and identify recurring insights.

The data collected in this study were subjected to certain statistical treatments.

The data will be coded, tallied, and tabulated for better presentation and interpretation of the results. These tools used to analyze the study will be in Microsoft Excel (PH-Stat). The statistical method that will be used is **W*eighted Mean.*** The weighted mean condition is a factual technique that computes the normal by duplicating the loads with their mean and taking their total. It is a kind of typical in which loads are doled out to individual qualities to decide the overall significance of every perception.

$$W = \frac{\sum_{i=1}^{n} w_i X_i}{\sum_{i=1}^{n} w_i}$$

W = weighted average

n = number of terms to be averaged

Wi = weights applied to x values

Xi = data values to be averaged

Table 3.8

Likert Scale

Numerical Value	Scale Ranges	Verbal Interpretation
5	4.50 - 5.00	Strongly Agree
4	3.50 - 4.49	Agree
3	2.50 – 3.49	Moderately Agree
2	1.50 – 2.49	Disagree
1	1.00 – 1.50	Strongly Disagree

The scale interval and its verbal interpretation according to the rating value with which the researcher can easily determine the respondents answer for determining the

level of user satisfaction and areas for improvement. The Numerical value of 5 has a scale range of 4.50 - 5.00 with a verbal interpretation of **Strongly Agree**, Numerical value 4 has a scale range of 3.50 - 4.49 with a verbal interpretation of **Agree,** Numerical value 3 has a scale range of 2.50 - 3.49 with a verbal interpretation of **Moderately Agree,** Numerical value 2 has a scale of 1.50 - 2.49 with a verbal interpretation of **Disagree,** Lastly, Numerical value 1 has a scale range of 1.00 - 1.50 with a verbal interpretation of **Strongly Agree.**

Chapter 4
RESULTS AND DISCUSSION

This chapter presents the PetEat system's project description, architecture, capabilities, limitations, and evaluation results. The outcomes of development and testing phases are discussed along with data collected during the system's user acceptance testing.

4.1 Project Description

PetEat is a comprehensive web and mobile-based system designed to streamline clinic operations, enhance pet healthcare services, and improve pet-owner communication through features such as NFC tracking, cloud-based EMRs, telemedicine readiness, appointment scheduling, and inventory management. Aligned with the project's specific objectives stated in Chapter 1, PetEat successfully implemented the following:

1. A Web & Mobile platform connecting pet owners and veterinary clinics.
2. Telemedicine readiness via a video consultation infrastructure.
3. Cloud-based EMR for digital medical records and automated vaccination

reminders.

4. NFC technology for rapid pet identification.
5. Inventory Management System to track medicine and supply stocks.

Functional Requirements:

- Multi-role authentication (Super Admin, Admin, Vet Clinic, Pet Owner)
- Pet registration and medical records management
- Appointment scheduling with time conflict resolution
- Digital prescriptions and inventory tracking
- Real-time activity feeds and notifications

To further support operational efficiency, the system implements dynamic form validation, role-based access controls, and automated email confirmations for registrations. These functionalities ensure that only authorized users can perform specific actions, thereby improving data integrity and security. Additionally, modular routing allows the platform to redirect users based on their roles, streamlining navigation and enhancing user experience for both veterinary professionals and pet owners.

Non-Functional Requirements:

- **Security**: JWT-based authentication
- **Scalability**: Modular and API-driven architecture
- **Usability**: Mobile-responsive, accessible design

To maintain quality of service, non-functional requirements also include cross-platform compatibility and responsive layout scaling to different screen sizes. The platform underwent responsive testing on multiple browsers and devices to guarantee consistent rendering across platforms. Additionally, to support long-term sustainability,

maintainability was emphasized 48 through modular code structure, reusable components, and proper documentation. These qualities make it easier for future developers to introduce improvements or fix issues with minimal disruption.

Development and Testing

- **Frontend**: Next.js 14, TypeScript, Tailwind CSS
- **Backend**: Node.js with Express.js
- **Database**: Firebase
- R**eal-Time Communication:** Socket.io
- **Authentication**: JWT-based
- **UI Components**: Shadcn/ui

During development, Agile sprints were used to iteratively build and test features, allowing quick adaptation based on team feedback. GitHub was utilized for version control and collaboration, ensuring code consistency and easier tracking of changes. Integration of development tools such as ESLint, Prettier, and Postman helped maintain clean code and standardized API testing. This collaborative and tool-driven approach played a vital role in delivering a reliable platform within the project timeline.

Testing procedures

- **Unit Testing**: Models, API endpoints, controllers
- **Integration Testing**: Full-stack workflow from UI to DB
- **UAT**: Conducted with staff from veterinary clinics

- **Security Testing**: Verified role-based access and secure data transmission

Regression testing was also integrated into the development cycle to ensure that new updates or feature enhancements did not inadvertently break existing functionality. Tools like Jest and Postman were used to automate and document tests for API endpoints, ensuring expected responses under various conditions. Additionally, test cases were developed, aligning technical validation and system use cases.

Implementation Plan

Before deployment, a pilot test plan is crafted to simulate real-world scenarios and user flows within a selected veterinary clinic. The plan includes walkthroughs for pet registration, appointment booking, NFC scanning, and EMR updating. Feedback sessions will be conducted to gather responses from both technical and non-technical users, which informed bug fixes and minor UI refinements. This phased implementation strategy helped mitigate potential issues and ensure a smoother transition to the live environment.

Deployment:

- Backend APIs deployed via **Render** or **AWS**
- Database hosted on **Firebase**
- Real-time features via **Socket.io**
- File management with **cloud storage integration**

The deployment strategy prioritizes uptime and ease of rollback, utilizing serverless architecture for the frontend and container-based deployment for the backend. Deployment scripts were written to automate builds and monitor health status using Render's observability tools and Firebase's real-time database monitoring. A staging environment will also be 50 maintained for testing patches before live deployment, reducing the risk of production failures and ensuring a continuous integration and delivery pipeline.

Implementation Results

The initial deployment of the PetEat platform successfully enabled users to:

- Manage pet profiles and medical records with cloud-based Electronic Medical Records (EMR).
- Schedule and manage veterinary appointments efficiently.
- Conduct telemedicine video consultations with veterinarians.
- Utilize NFC technology for quick identification and lost pet reporting.
- Track and manage inventory for veterinary clinics

Pet owners and veterinary clinic staff reported positive experiences with the streamlined workflows, and improved access to pet healthcare services. Administrators were effectively able to manage user profiles, and monitor overall system activity. While PetEat achieved its key goals, some limitations remain, such as lack of integrated payment gateway, multi-language or accessibility settings, and reliance on stable internet access in remote areas, detailed in Section 4.3

4.2 Project Structure

The system's architecture was purposefully divided into distinct layers—presentation, application, and data—to allow for easier debugging, scalability, and future feature integrations. Each layer is responsible for a specific set of tasks, thereby enforcing a clear separation of concerns. This structural clarity facilitates onboarding new developers, supports integration, and simplifies platform maintenance or upgrades without affecting the user experience.

Frontend Design (Next.js)

- **Authentication Module**: Public and private route validation via JWT
- **Dashboards**: Role-specific interfaces for Clinic, Pet Owners, Admin, Super Admin
- **Pet Module**: CRUD for pet profiles and medical records
- **Appointments**: Time-slot booking and status tracking
- **Inventory & Prescriptions**: Stock updates, medicine assignment

The frontend was optimized not just for functionality but also for accessibility and responsiveness. Tailwind CSS was used to create adaptive UI components, ensuring usability across varying screen resolutions, from mobile phones to desktop browsers. Features such as real-time form feedback, and error state indicators were also incorporated based on early user testing. These design enhancements collectively aim to create a more inclusive and intuitive user experience.

Backend Architecture (Node.js/Express.js)

- **REST API** structure with secure endpoints
- **Controllers** for Pets, Users, Appointments, Inventory
- **Socket Handler**: Broadcast updates across connected clients
- **Middleware**: For authentication and role-checking

The backend leverages Express.js middleware to handle authentication, logging, and error management efficiently. Routes are grouped according to resource types (e.g., pets, users, appointments) and employ RESTful conventions, enabling consistent and scalable API integration. Token expiration, refresh mechanisms, and role checks are implemented to secure endpoints, while caching strategies and data pagination were added to ensure optimal performance, especially during high-traffic operations.

Database (Firebase)

- **Users**: Roles and profile data
- **Pets**: Health records, ownership mapping
- **Appointments**: Doctor-patient linkage
- **Prescriptions/Inventory**: Linked by medicineID and clinicID

Firebase's real-time database enables asynchronous data access and updates, which is essential for modules like appointments, chat, and telemedicine. Data is normalized to reduce redundancy, and indexing is used to speed up search queries such as pet lookups or user histories. To enhance data security, sensitive information is

encrypted both at rest and in transit, while Firebase rules restrict access based on authentication states and user roles.

4.3 Project Capabilities and Limitations

The platform's current architecture ensures a strong foundation for future enhancements. However, due to budget and time constraints, advanced analytics dashboards and offline mobile support could not be fully implemented during this phase. These features 53 will be possible implementations for the next iteration of development, pending user demand and funding.

System Capabilities

- Real-time dashboard analytics
- Multi-role access control
- Appointment, EMR, prescription, and inventory modules
- Secure login with JWT
- Real-time updates via WebSockets
- Cloud-based data storage and access

The system is also equipped to handle concurrent user sessions, allowing multiple pet owners and veterinarians to interact with the platform simultaneously without performance degradation. With its modular build and real-time communication features, PetEat can scale to support additional veterinary branches, regional databases, or cloud-hosted services in the future.

System Limitations

- No payment gateway integration yet
- No multi-language or accessibility settings for special users

Another limitation involves the system's reliance on stable internet connectivity, which can be problematic in rural areas. Since key modules like telemedicine and cloud storage are dependent on real-time data transmission, intermittent connections can disrupt service 54 continuity. Additionally, the current system lacks localized language support, which may hinder usability for non-English-speaking users. Addressing these limitations will be critical in scaling the platform to broader demographics.

4.4 Project Evaluation

It presents all statistical results/findings based on the collected responses from the respondents. The results may be presented in the form of tables or graphs with corresponding analysis. The evaluation procedure utilized the ISO 25010:2011 Software Product Quality Standards, assessing Functional Suitability, Performance Efficiency, and Usability. The Likert scale, ranging from 1 (Strongly Disagree) to 5 (Strongly Agree), was used for quantitative assessment. The data was processed using weighted mean calculations, as defined in Chapter 3.

Table 4.1

Results of Functional Suitability - represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions (IT Experts)

Functional Suitability	**Mean**	**Verbal Interpretation**
Functional Completeness	3.9	Agree
Functional Correctness	4.5	Strongly Agree
Functional Appropriateness	4.4	Agree
Overall Weighted Mean	**4.27**	**Agree**

This table presents the mean and the interpretation of the Functional Suitability. As shown in the table, the majority of the IT Experts Agree with an overall weighted mean of 4.27.

Table 4.2

Results of Performance Efficiency - represents the performance relative to the amount of resources used under stated conditions. (IT Experts)

Performance Efficiency	**Mean**	**Verbal Interpretation**
Time Behavior	4	Agree
Resource Utilization	4.4	Agree
Capacity	4.3	Agree
Overall Weighted Mean	**4.23**	**Agree**

This table presents the mean and the interpretation of the Performance Efficiency. As shown in the table, the majority of the IT Experts Agree with an overall weighted mean of 4.23.

Table 4.3

Results of Usability - Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use. (IT Experts)

Usability	**Mean**	**Verbal Interpretation**
Appropriateness Recognizability	4.3	Agree
Learnability	4.6	Strongly Agree
Operability	4.1	Agree
User Error Protection	4.4	Agree
User Interface Aesthetics	4	Agree
Accessibility	4.4	Agree
Overall Weighted Mean	**4.3**	**Agree**

This table presents the mean and the interpretation of the Usability. As shown in the table, the majority of the IT Experts Agree with an overall weighted mean of 4.3.

Table 4.4

Results of Functional Suitability - represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions (End Users)

Functional Suitability	**Mean**	**Verbal Interpretation**
Functional Completeness	4.1	Agree
Functional Correctness	4.3	Agree
Functional Appropriateness	4.43	Agree
Overall Weighted Mean	**4.27**	**Agree**

This table presents the mean and the interpretation of the Functional Suitability. As shown in the table, the majority of the End Users Agree with an overall weighted mean of 4.27.

Table 4.5

Results of Performance Efficiency - represents the performance relative to the amount of resources used under stated conditions. (End Users)

Performance Efficiency	**Mean**	**Verbal Interpretation**
Time Behavior	4.1	Agree
Resource Utilization	4.27	Agree
Capacity	4.07	Agree
Overall Weighted Mean	**4.15**	**Agree**

This table presents the mean and the interpretation of the Performance Efficiency. As shown in the table, the majority of the End Users Agree with an overall weighted mean of 4.15.

Table 4.6

Results of Usability - Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use. (IEnd Users)

Usability	**Mean**	**Verbal Interpretation**
Appropriateness Recognizability	4.23	Agree
Learnability	4.5	Strongly Agree
Operability	3.9	Agree
User Error Protection	4.1	Agree
User Interface Aesthetics	4.3	Agree
Accessibility	4.43	Agree
Overall Weighted Mean	**4.24**	**Agree**

This table presents the mean and the interpretation of the Usability. As shown in the table, the majority of the IT Experts Agree with an overall weighted mean of 4.24.

Chapter 5

FINDINGS, CONCLUSION AND RECOMMENDATION

This section discusses the most significant findings, interrelations of all the variables of the study, and specific recommendations for further research or practical steps that can be taken.

5.1 Summary of Findings

The PetEat platform has received strong positive feedback from both end-users (veterinarians and pet owners) and IT experts, demonstrating high satisfaction. The overall weighted means fall under the "Agree" interpretation, with specific attributes like Functional Completeness, Learnability, and User Error Protection consistently achieving “Agree" ratings. This highlights the system's comprehensive features, ease of use, and robust error protection. While scores for Operability (end-users) and Capacity (end-users) and Time Behaviour and Functional Completeness (IT experts) were slightly lower yet still positive, they indicate areas where future performance enhancements and infrastructure optimization could further improve the user experience, especially considering the platform's reliance on stable internet connectivity.

5.2 Conclusions

Based on the findings, the PetEat platform stands as an effective and well-received improvement for modernizing veterinary care and pet ownership. The study concludes that the platform successfully addresses the central problem of fragmented and inefficient pet care management by integrating critical functionalities—telemedicine, cloud-based EMR, NFC tracking, and inventory management into a single, accessible system. This integration has been validated through overwhelmingly positive feedback from both IT experts and end-users, demonstrating its functional suitability, usability, and performance efficiency. This study's outcomes contribute to the growing body of literature on digital transformation in healthcare, providing a practical model applicable to wider and similar populations facing challenges in localized, specialized care. The consistent positive evaluation, especially regarding usability and error protection, implies that investing in user-centric design within technological solutions can lead to high user 61 adoption and satisfaction, driving broader societal benefits in animal welfare and public health.

5.3 Recommendations

1. **For End-Users (Pet Owners and Veterinarians):** We encourage all pet owners and veterinary clinics to explore and use every part of the PetEat platform. Regularly updating pet profiles, booking appointments through the system, and using telemedicine will make managing pet care much smoother and more efficient. Please continue to give the researchers your feedback. Your suggestions help us understand

what works well and what can be made even better, ensuring PetEat truly serves your needs

2. **For Future Research and Development:** Future development should focus on adding a secure way for users to make payments directly within the app for consultations or other services. This will make transactions easier for everyone. Develop more detailed reports and dashboards for veterinary clinics. This will help them easily see trends in appointments, popular services, inventory and especially to keep track of pet sickness for decision making of the veterinary clinics. Consider adding support for different languages to make PetEat accessible to more users, especially in diverse communities. Also, explore features that help users with specific accessibility needs. Investigate ways for the mobile app to work even without an internet connection, at least for basic information like stored medical records. This would be very helpful in areas where internet access is unreliable. For PetEat to serve a very large number of users without slowing down, look into technologies that help distribute the workload. This will ensure the app stays fast and responsive even when many people are using it at once. Conduct follow-up studies to see how PetEat benefits users and clinics over a longer period. This will help understand its lasting positive effects on pet care. Implement RFID tags to further consolidate tracking of pets and utilize realtime Tracking.

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APPENDICES A

ISO-25010-Project Evaluation Questionnaire

Dear Respondent,

Thank you very much for using **PetEat: A Web & Mobile Platform for Veterinary Clinics and Pet Lovers with NFC Tracking, Telemedicine, and Cloud Based Electronic Medical Record**. Give brief details about your capstone project, example – This capstone project is a Database management system for e-Governance solution that provides a systematic approach in managing the residents profile and enables document generation for effective planning and monitoring at the barangay level.

In this regard, we would like to seek your cooperation by accomplishing this survey form to help us evaluate the quality and assess the level of acceptability of the application to you as our target user/partner. Your feedback will be used to further improve its features to serve you better. Rest assured that the data gathered through this survey will be treated with utmost confidentiality in accordance to the provisions of RA 10173 or the Data Privacy Act of 2012.

Again, thank you very much for being our partner on this endeavor.

☐ End Users (Veterinarians, Pet Owners)

☐ IT - Experts

Name: _____
Age: _____
Affiliated Institution: _____
Position/Designation: _____

PART 1: SOFTWARE QUALITY EVALUATION

A. Directions: Please evaluate the software quality of the "**PetEat: A Web & Mobile Platform for Veterinary Clinics and Pet Lovers with NFC Tracking, Telemedicine, and Cloud Based Electronic Medical Record"**based on the three characteristics of ISO 25010 Software Product Quality Standards for end user. Choose on the appropriate box using the four-point scale with its corresponding verbal descriptions (4 as the highest and 1 as the lowest).

Functional Suitability - represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
Functional completeness - Degree to which the set of functions covers all the specified tasks and user objectives					
Functional correctness - Degree to which a product or system					

provides the correct results with the needed degree of precision.					
Functional appropriateness - Degree to which the functions facilitate the accomplishment of specified tasks and objectives.					

Performance Efficiency - represents the performance relative to the amount of resources used under stated conditions.

Time behaviour - Degree to which the response and processing times and throughput rates of a system, when performing its functions, meet requirements.					
Resource utilization - Degree to which the amounts and types of resources used by a system, when performing its functions, meet requirements.					
Capacity - Degree to which the maximum limits of a product or system parameter meet requirements.					

Usability - Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use.

Appropriateness recognizability - Degree to which users can recognize whether a system is appropriate for their needs.					
Learnability - Degree to which a system can be used by specified users to achieve specified goals of learning to use the product or system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.					
Operability - Degree to which a system has attributes that make it easy to operate and control.					

User error protection - Degree to which a system protects users against making errors.					
User interface aesthetics - Degree to which a user interface enables pleasing and satisfying interaction for the user.					
Accessibility - Degree to which a system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.					

Respondent's Signature: _____

Date of Evaluation: _____

COMMUNICATION LETTERS

THESIS READER CERTIFICATION

RESEARCHER PROFILE

Deiniel Banaticla

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Skills

Editing

Coding

Leadership

Creativity

Flexibility

About

A dedicated College Student with maintained high grades and willing to learn and develop skills. Striving his goal to make the world and himself better.

Education History

Saint Michael's College of Laguna

- Grade 10 (SY 2019-2020) - Rank 6th among the class
- Grade 11 (SY 2020-2021) - With Honors
- Grade 12 (SY 2021-2022) - With Honors

FEU ALABANG

- Bachelor of Science in Information Technology Specialization in Web and Mobile Application

Objective

- To secure a challenging position in a reputable organization to expand my learnings, knowledge, and skills.
- To graduate as Bachelor of Science in Information Technology

Other Skills

- Typing Skills
- Decision Making
- Multitasking

Certification

- Certificate of Recognition (2020)
- With Honors (2021-2022)
- Certification exam of Information Technology Specialist in Networking (2024)
- Certificate of Recognition in Design Thinking (1st runner up) (2024)

Experience

- Have a experience at Computer Programming Since High School
- Experienced editing different kinds
- Participated in contests at school such as quiz and spelling bee etc.

Hobbies and Interests

- Gaming
- Travel
- Listening to Music
- Sports

Sean Perez

COLLEGE STUDENT

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Skills

Editing

Coding

Leadership

Creativity

Flexibility

About

A dedicated College Student with maintained high grades and willing to learn and develop skills. Eager to learn new things

Education History

San Beda College Alabang - Manresa School

- Grade 9 (SY 2018 - 2019) -4th Place Photoshop Competition
- Grade 10 (SY 2019 - 2020) - CAT Officer (Flight Leader Alpha)

FEU ALABANG

- Grade 11 (SY 2020-2021) - With Honors
- Grade 12 (SY 2021-2022) - With Honors
- Bachelor of Science in Information Technology Specialization in Web and Mobile Application

Objective

- To secure a challenging position in a reputable organization to expand my learnings, knowledge, and skills.
- To graduate as Bachelor of Science in Information Technology

Other Skills

- Multitasking
- Critical Thinker
- Adaptive

Certification

- With Honors (2021-2022)
- Certification exam of Information Technology Specialist in Networking (2024)
- Certificate of Recognition in Design Thinking (1st runner up) (2024)

Hobbies and Interests

- Gaming
- Travel
- Playing instruments
- Reading
- Working Out

Lee Sabroso

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Skills

Editing

Coding

Leadership

Creativity

Flexibility

About

A dedicated College Student with maintained high grades and willing to learn and develop skills. Striving his goal to make the world and himself better.

Education History

FEU Alabang

- Grade 11 (SY 2020-2021) - With Honors
- Grade 12 (SY 2021-2022) - With Honors

FEU ALABANG

- Bachelor of Science in Information Technology Specialization in Web and Mobile Application

Objective

- To secure a challenging position in a reputable organization to expand my learnings, knowledge, and skills.
- To graduate as Bachelor of Science in Information Technology

Other Skills

- Typing Skills
- Decision Making
- Multitasking

Certification

- With Honors (2021-2022)
- Certification exam of Information Technology Specialist in Networking (2024)
- Certificate of Recognition in Design Thinking (1st runner up) (2024)

Experience

- Have a experience at Computer Programming Since High School
- Experienced editing different kinds
- Participated in contests at school such as quiz and spelling bee etc.

Hobbies and Interests

- Gaming
- Travel
- Biking

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Skills

Editing

Coding

Leadership

Creativity

Flexibility

About

A college student with an ambition to become a professional IT, and acquiring the needed skills to be a good employee in a corporate environment.

Education History

St. Anthony, St. Francis of Assisi, LPENHS-TVA

- Gr.8 - Spelling Bee Gold Medalist
- Gr. 10 - Math Quiz Bee Silver Medalist
- Gr. 10 - With Honors

FEU ALABANG

- Gr.11 - With Honors
- Gr. 12 - With Honors
- Bachelor of Science in Information Technology 2022-2026

Objective

- To secure a challenging position in a reputable organization to expand my learnings, knowledge, and skills.
- To graduate as Bachelor of Science in Information Technology

Other Skills

- Typing Skills
- Decision Making
- Multitasking

Certification

- Certificate of Recognition (2018)
- With Honors (2019-2021)
- Certification exam of Information Technology Specialist in Networking (2024)
- Certificate of Recognition in Design Thinking (1st runner up) (2024)

Experience

- McDonald's Server and Janitor (2020)
- Work Immersion in Civil Engineering (2020)

Hobbies and Interests

- Gaming
- Meditating
- Running